DO YOU KNOW?

Melatonin is secreted at night, Light falling on the retina of eye decreases melatonin production. Serotonin is secreted during day time.

Circadian rhythm is the 24 hour cycle of biological activities associated with natural periods of light and darkness. Example sleep wake cycle, body temperature, appetite etc.

11.2.4 Thyroid gland

The butterfly shaped thyroid gland is a bilobed gland located below the larynx on each side of upper trachea. It is the largest endocrine gland in the body. Its two lateral lobes are connected by a median tissue mass called **isthmus**. Each lobe is made up of many lobules .The lobules consist of follicles called **acini (acinus in singular)**. Each acinus is lined with glandular, cuboidal or squamous epithelial cells. The lumen of acinus is filled with colloid, a thick glycoprotein mixture consisting of thyroglobulin molecules.

Hormones of the thyroid gland are often called the major metabolic hormones. The follicular cells of thyroid gland secrete two hormones namely tri-iodothyronine (T_3) and thyroxine or tetra- iodothyronine (T_4). The parafollicular cells or 'C' cells of thyroid gland secrete a hormone called thyrocalcitonin. Iodine is essential for the normal synthesis of thyroid hormones. Thyroid releasing hormone from the hypothalamus stimulates the adenohypophysis to secrete TSH, which inturn stimulates the thyroid gland to secrete the thyroid hormones. Thyroid hormones show a negative feedback effect on the hypothalamus and pituitary (Figure11. 4).

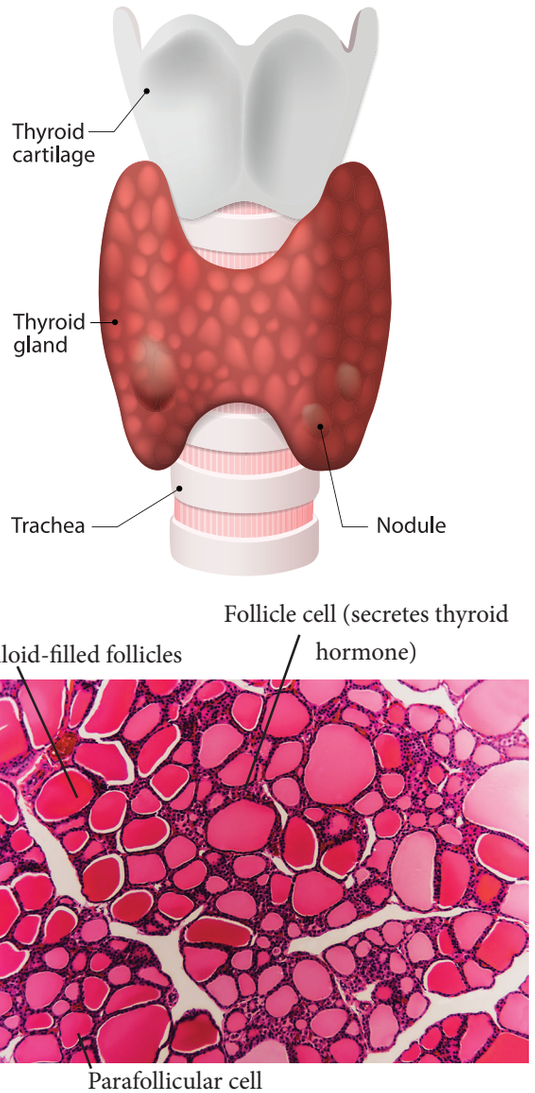


Figure 11. 3 Structures of thyroid gland

Functions of thyroxine or tetra-iodothyronine (T4): Thyroxine regulates the basal metabolic rate (BMR) and body heat production. It stimulates protein synthesis and promotes growth. It is essential for the development of skeletal and nervous system. Thyroxine plays an important role in maintaining blood pressure. It reduces serum cholesterol levels, Optimum levels of thyroxine in blood is necessary for gonadial functions.

Sporodic goitre is a genetic disease and is not caused by iodine or thyroxine deficiency.

XI Std Zoology Chapter-11 EM.indd 208 30/12/2022 14:00:51